Reduction Modulo p

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1. Motivation and reduction modulo p

In characteristic zero, singularities can be studied through resolutions $Y \to X$; in characteristic $p > 0$, the Frobenius endomorphism $F(r) = r^p$ gives a different set of tools. Reduction modulo p links these approaches.

Let R be a finitely generated algebra over a characteristic-zero field K. Choose a finitely generated $\mathbb{Z}$-subalgebra $A \subset K$ and a finite-type A-algebra R_A satisfying $R_A \otimes_A K \cong R$. For a closed point $s \in \operatorname{Spec} A$ with residue field $\kappa(s)$ of characteristic $p > 0$, set

$$R_s = R_A \otimes_A \kappa(s).$$

For a property P of positive-characteristic rings, R has *dense P-type* if R_s has P for a Zariski-dense set of closed points, and *open P-type* if it holds for every closed point in some nonempty open subset of $\operatorname{Spec} A$.

Classical properties. After choosing a suitable model and imposing the necessary finiteness or flatness hypotheses, one can compare the following properties with those of the fibers: *reducedness, regularity, normality, Cohen–Macaulayness, Serre's S_2 condition, Gorensteinness,* and *complete-intersection property.* Thus the method concerns ordinary algebraic properties as well as singularities. These comparisons require a well-behaved model: in practice one often first localizes A and uses generic freeness to arrange flatness.

2. Frobenius singularities

Let R be Noetherian of characteristic $p > 0$. The Frobenius-twisted module $F^e_* R$ has scalar action $a \cdot F^e_* b = F^e_*(a^{p^e} b)$.

- **F-finite:** $F_* R$ is a finite R-module.
- **Strongly F-regular (SFR):** for every c outside the minimal primes, some map $R \to F^e_* R$, $1 \mapsto F^e_*(c)$, splits.
- **F-rational:** all parameter ideals are tightly closed.
- **F-injective:** Frobenius acts injectively on each $H^i_{\mathfrak{m}}(R)$ for every local ring $(R, \mathfrak{m})$.

For an ideal I and $q = p^e$, put $I^{[q]} = (f^q : f \in I)$. An element x is in the *tight closure* I^* if there is a c outside the minimal primes with $cx^q \in I^{[q]}$ for every sufficiently large q. Thus F-rational means $I = I^*$ for every parameter ideal. In the usual F-finite setting, strong F-regularity implies F-rationality and F-purity; furthermore F-splitting and F-purity coincide.

3. Comparison with characteristic-zero singularities

The following correspondences were highlighted (with the usual hypotheses on the algebra and singularity class):

Characteristic zero	**Reduction mod p**
Kawamata log terminal (klt)	open strongly F-regular type
Rational singularities	dense/open F-rational type
Du Bois singularities	implied by dense F-injective type

In the first two rows the intended relationships are equivalences under the standard hypotheses (Hara–Watanabe, Smith, Hara). In the third row, the established direction is

$$\text{dense } F\text{-injective type} \implies \text{Du Bois singularities.}$$

The converse is still an open *question*: does every Du Bois singularity have dense F-injective type? The comparison relies on deeper techniques, including derived methods mentioned in the seminar.

4. Frobenius on local cohomology and Fedder's criterion

Take a perfect field k of characteristic $p > 0$ and put

$$S = k[x_1, \dots, x_n]_{(x_1,\dots,x_n)}, \quad \mathfrak{n} = (x_1, \dots, x_n)S, \quad R = S/(g), \quad \mathfrak{m} = \mathfrak{n}R,$$

where $0 \neq g \in \mathfrak{n}$. The short exact sequence $0 \to S \xrightarrow{g} S \to R \to 0$ yields

$$H^{n-1}_{\mathfrak{m}}(R) \cong (0 :_{H^n_{\mathfrak{n}}(S)} g).$$

Write $\eta = [1/(x_1 \cdots x_n)]$ for the socle generator in $H^n_{\mathfrak{n}}(S)$, represented using the Čech description

$$H^n_{\mathfrak{n}}(S) \cong \frac{S_{x_1 \cdots x_n}}{\sum_{i=1}^n S_{x_1 \cdots \widehat{x_i} \cdots x_n}}.$$

Under the above identification, Frobenius acts through $g^{p-1}F_S$; hence

$$F_R(\eta) = g^{p-1}F_S(\eta) = \left[\frac{g^{p-1}}{x_1^p \cdots x_n^p}\right].$$

This class is nonzero precisely when $g^{p-1} \notin \mathfrak{n}^{[p]} = (x_1^p, \dots, x_n^p)S$. For the hypersurface (which is Gorenstein), Frobenius injectivity on top local cohomology detects F-purity. Thus

$$\boxed{S/(g) \text{ is } F\text{-pure} \iff g^{p-1} \notin \mathfrak{n}^{[p]}.}$$

More generally, for $R = S/I$ with S regular local and F-finite, *Fedder's criterion* is

$$S/I \text{ is } F\text{-pure} \iff (I^{[p]} : I) \not\subseteq \mathfrak{n}^{[p]}.$$

For $I = (g)$, we have $(I^{[p]} : I) = (g^{p-1})$, recovering the hypersurface case.